

3

Simplify.

A

$$22ay \div 2ay = \underline{\hspace{2cm}}$$

B

$$12xy \div 4xy = \underline{\hspace{2cm}}$$

C

$$25ab \div 5ab = \underline{\hspace{2cm}}$$

D

$$8xy \div xy = \underline{\hspace{2cm}}$$

E

$$\frac{14km}{7km} = \underline{\hspace{2cm}}$$

F

$$\frac{16ab}{32ab} = \underline{\hspace{2cm}}$$

G

$$-25xy \div 5x = \underline{\hspace{2cm}}$$

H

$$-12m \div (-3mn) = \underline{\hspace{2cm}}$$

I

$$6abc \div 2ab = \underline{\hspace{2cm}}$$

J

$$ab \div 3ab = \underline{\hspace{2cm}}$$

K

$$36kmn \div 6mn = \underline{\hspace{2cm}}$$

L

$$4abc \div 20abc = \underline{\hspace{2cm}}$$

M

$$-18m \div (-9k) = \underline{\hspace{2cm}}$$

N

$$4ab \div (-12a) = \underline{\hspace{2cm}}$$

O

$$\frac{-9xy}{12xy} = \underline{\hspace{2cm}}$$

P

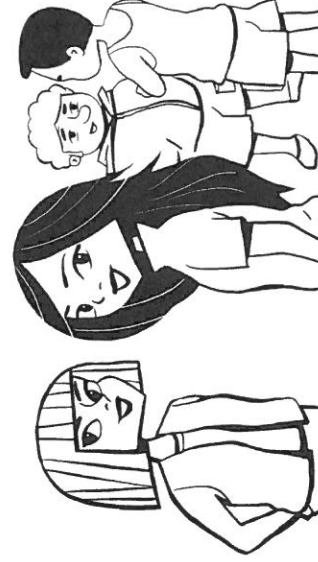
$$\frac{12mn}{-3n} = \underline{\hspace{2cm}}$$

Q

$$\frac{-20km}{-5mn} = \underline{\hspace{2cm}}$$

R

$$\frac{-8abc}{16bc} = \underline{\hspace{2cm}}$$



Remember:

$$a^2 = a \times a \text{ so } a^2 \div a = a$$

4

Simplify.

A

$$3a^2 \div a = \underline{\hspace{2cm}}$$

B

$$64y^2 \div 8 = \underline{\hspace{2cm}}$$

C

$$10n^2 \div n^2 = \underline{\hspace{2cm}}$$

D

$$6x^2 \div 3x = \underline{\hspace{2cm}}$$

E

$$12n^2 \div 2n = \underline{\hspace{2cm}}$$

F

$$56k^2 \div 7k^2 = \underline{\hspace{2cm}}$$

G

$$\frac{2a^2}{a} = \underline{\hspace{2cm}}$$

H

$$\frac{10n^2}{2n} = \underline{\hspace{2cm}}$$

I

$$\frac{6x^2y}{3x} = \underline{\hspace{2cm}}$$

J

$$\frac{15ab^2}{5b} = \underline{\hspace{2cm}}$$

K

$$\frac{5a^2}{15ab} = \underline{\hspace{2cm}}$$

L

$$\frac{a^2b}{ab^2} = \underline{\hspace{2cm}}$$

M

$$\frac{4a^2b}{16ab^2} = \underline{\hspace{2cm}}$$

N

$$\frac{3kn^2}{12k^2n^2} = \underline{\hspace{2cm}}$$

O

$$20a^2b \div (-10ab) = \underline{\hspace{2cm}}$$

P

$$-15kn \div 5k^2 = \underline{\hspace{2cm}}$$

Q

$$-12xy^2 \div (-4xy) = \underline{\hspace{2cm}}$$

R

$$-8ab \div (-10a^2) = \underline{\hspace{2cm}}$$



Simplifying expressions by division 2A

ALGEBRAIC EXPRESSIONS

UNIT 2
TOPIC 4

A division can be written as a fraction.

For example, $15kn \div 5n$ can be written as $\frac{15kn}{5n}$.

Sometimes you cannot complete the division.

There may be numbers or pronumerals left in the denominator.

When this happens, leave your answer as a fraction.

1 Simplify.

A $\frac{10x}{2} = \underline{\hspace{2cm}}$

B $\frac{14a}{7} = \underline{\hspace{2cm}}$

C $\frac{xy}{x} = \underline{\hspace{2cm}}$

D $\frac{3ab}{b} = \underline{\hspace{2cm}}$

E $\frac{5ad}{a} = \underline{\hspace{2cm}}$

F $\frac{8kn}{kn} = \underline{\hspace{2cm}}$

G $\frac{12a}{3a} = \underline{\hspace{2cm}}$

H $\frac{25xy}{5x} = \underline{\hspace{2cm}}$

I $\frac{30mn}{6m} = \underline{\hspace{2cm}}$

J $\frac{21ab}{3b} = \underline{\hspace{2cm}}$

K $\frac{-18fg}{9g} = \underline{\hspace{2cm}}$

L $\frac{12kt}{-4k} = \underline{\hspace{2cm}}$



2 Simplify.

A $\frac{2a}{10} = \underline{\hspace{2cm}}$

B $\frac{3y}{12} = \underline{\hspace{2cm}}$

C $\frac{10x}{8} = \underline{\hspace{2cm}}$

D $\frac{15m}{10} = \underline{\hspace{2cm}}$

E $\frac{10k}{25} = \underline{\hspace{2cm}}$

F $\frac{6a}{12a} = \underline{\hspace{2cm}}$

G $\frac{3a}{9y} = \underline{\hspace{2cm}}$

H $\frac{8a}{6b} = \underline{\hspace{2cm}}$

I $\frac{3kn}{6k} = \underline{\hspace{2cm}}$

J $\frac{5a}{10b} = \underline{\hspace{2cm}}$

K $\frac{8x}{xy} = \underline{\hspace{2cm}}$

L $\frac{15a}{20ab} = \underline{\hspace{2cm}}$

M $\frac{abc}{abd} = \underline{\hspace{2cm}}$

N $\frac{kmn}{2kn} = \underline{\hspace{2cm}}$

O $\frac{abc}{3ac} = \underline{\hspace{2cm}}$

P $\frac{2km}{5mn} = \underline{\hspace{2cm}}$

